<u>Question Bank</u>

1. Describe the process of frequency domain sampling and reconstruction of discrete time signals.
2. Define DFT and IDFT of a signal Show DFT and IDFT as a linear transformation.
3. State and prove the following basic properties of DFT (i) Periodicity (ii) Linearity (iii) Symmetry property for a real valued sequence.
4. State and prove the following properties (i) Time reversal property (ii) Circular time shift Property (iii) Circular Frequency shift Property (iv) Modulation Property (v) Complex Conjugate Property (
5. State and prove Parseval's Energy theorem.
6. Compute the N point DFT of the sequence $x(n) = a^n$; $0 \leq n \leq (N - 1)$
7. . Find the DFT of the sequence $x(n) = \begin{pmatrix} 1; & 0 \leq n \leq 2 \\ 0; & otherwise \end{pmatrix}$ for (i) N = 4 (ii) N = 8 . Plot magnitude spectrum of X(K) and comment on the result obtained.
8. Compute the N – Point DFT of the sequence $x(n) = \frac{1}{2} + \frac{1}{2}\cos\left[\frac{2\pi}{N}\left(n - \frac{N}{2}\right)\right]$
9. Find the 5 point of the discrete time signal x(n) = { 1, 2, 3, 1 }
10. Find the IDFT of X(k) = { 4, – 2j, 0, 2j }
11. Given X(k) = { 2, 1 + j, 0, 1 – j } find x(n)
12. Solve by concentric circles or graphical method to find the circular convolution x(n) = { 1, 3, 5, 3}, h(n) = {2,3,1,1}
13. Compute 4 point circular convolution of the sequences using time domain and frequency domain x(n) = {2, 1, 2, 1} ; h(n) = { 1, 2, 3, 4**}**
14. If x(n) = { 1, 2, 1, –1, 2, 3 } Determine (i) $x((n - 3))_6$ (ii) $x((n + 2))_6$ and (iii) $x((-n))_6$
15. Determine the 4 point DFT of the sequence $x(n) = \cos\left(\frac{\pi n}{4}\right) + \sin\left(\frac{\pi n}{4}\right)$ using linearity property
16. The first 5 samples of 8 point DFT X(k) are given as follows: X(0) = 0.25, X(1) = 0.125 – j 0.318, X(4) = X(6) = 0 and X(5) = 0.125 – j 0.518. Determine the remaining samples, if x(n) is a real valued sequence
17. Find Y(k) if $x_1(n) = 1$; $0 \leq n \leq 7$, $x_2(n) = \cos(\pi n/4)$; $0 \leq n \leq 7$ and $y(n) = x_1(n).\ x_2(n)$
18. If $h(n) = \begin{bmatrix} \frac{1}{2}; & -2 \leq n \leq 2 \\ 0; & \text{otherwise} \end{bmatrix}$ Determine the 8 point DFT of h(n).

19. If $x(n) = 4\delta(n) + 3\delta(n-1) + 2\delta(n-2) + \delta(n-3)$; Compute the 6 point DFT of the signal x(n) and also determine finite length sequence y(n) such that $Y(k) = W_6^{4k} X(k)$

20. Using time shifting property, find the 8 point DFT of the sequence
$x(n) = \delta(n) + \delta(n-4) + 2\delta(n-5)$

21. If x(n) be a finite length sequence with X(k) = { 10, 1 – j, 4, 1 + j }; Using the properties of the DFT, find the DFT of the following (i) $x_1(n) = e^{j\pi n/2}\, x(n)$ (ii) $x_2(n) = \cos(\pi n/2)\, x(n)$ (iii) $x_3(n) = x((n-1))_4$

22. The 5 point DFT's of a complex sequence x(n) are { j, (1+j), (1+j2), (2+j2), (4+ j) }.
Compute Y(k) if $y(n) = x^*(n)$

23. For x(n) = { 1, - 2, 3, - 4, 5, - 6 }, without computing DFT, find the following

$$\text{(i)}X[0] \quad \text{(ii)}X[3] \quad \text{(iii)} \sum_{0}^{5} X[k] \quad \text{(iv)} \sum_{0}^{5} |X[k]|^2 \quad \text{(v)} \sum_{0}^{5} (-1)^k X[k]$$

24. A FIR filter has an impulse response of h(n)={1,2,3}. Determine the response of the filter to the input sequence x(n)= { 1, 2 }. Use DFT and IDFT. Verify the result using direct computation of linear convolution.

25. Consider the FIR filter with impulse response h(n) = { 3, 2, 1,1}. If the input is
x(n) = { 1, 2, 3, 3, 2, 1, -1, -2, -3, 5, 6, -1, 2, 0, 2, 1 }. Find the output. Use (i) overlap – save method.(ii) Overlap Add method   Assume the length of the block = 6